Year 10 Mathematics

Number 1

1.3 Significant Figures and Decimal Places

Significant Figures and Decimal Places

STARTER · 5 MIN

Date: _____ **Title:** Significant Figures and Decimal Places

LEARNING OBJECTIVES

- Round numbers to a given number of decimal places
- Round numbers to a given number of significant figures
- Estimate answers by rounding each number to 1 significant figure

STARTER *Answer in your book. No calculators.*

1. Round 367 to the nearest 10
2. Round 2649 to the nearest 1000
3. Round 5.83 to the nearest whole number
4. Which is bigger: 0.35 or 0.4?

Starter — Answers

STARTER

Tick or correct in green pen

1. 370
2. 3000
3. 6
4. 0.4 (0.40 > 0.35)

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can round numbers to a given number of decimal places	☐
SILVER	I can round numbers to a given number of significant figures	☐
GOLD	I can estimate answers by rounding each number to 1 significant figure	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Rounding

Look at the next digit: 5 or more → round up. 4 or less → the digit stays the same.

Decimal places (d.p.)

Count the digits after the decimal point.

7.1361 → 7.14 (2 d.p.)

Significant figures (s.f.)

Start counting at the first digit that is not zero.

0.001349 → 0.0013 (2 s.f.)

Keep the size of the number

Fill with zeros up to the decimal point.

87 602 → 90 000 (1 s.f.)

Estimating

Round every number to 1 s.f., then work it out.

$$\frac{39.8 \times 5.1}{0.49} \approx \frac{40 \times 5}{0.5} = 400$$

Examples 1 and 2 — Decimal Places

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Write 3.47 correct to 1 decimal place.

Example 2

Write 19.997 correct to 2 decimal places.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

3.4 | 7

The next digit is 7.

7 is 5 or more, so round up.

= **3.5**

Tip: draw a line after the digits you keep.

Example 2

19.99 | 7

The next digit is 7, so round up.

19.99 → 20.00

= **20.00**

Tip: 2 d.p. needs 2 digits after the point.

Task 1

PRACTICE · 9 MIN

BRONZE *Round each number*

1. Write 5.8394 correct to 2 d.p.

2. Write 34.777 correct to 2 d.p.

3. Write 3.009 correct to 1 d.p.

4. Write 9.96 correct to 1 d.p.

5. Write 0.0349 correct to 2 d.p.

6. Write 12.4951 correct to 3 d.p.

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. 5.84
2. 34.78
3. 3.0
4. 10.0
5. 0.03
6. 12.495

Examples 3 and 4 — Significant Figures

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Write 1361 correct to 2 significant figures.

Example 4

Write 0.0349 correct to 2 significant figures.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

13 | 61

The next digit is 6, so round up.

= **1400**

Tip: 1361 is about 1400, not 14.

Example 4

0.034 | 9

The first significant figure is 3.

The next digit is 9, so round up.

= **0.035**

Tip: zeros at the front are not counted.

Task 2

PRACTICE · 9 MIN

SILVER *Round each number*

1. Write 783 correct to 1 s.f.
2. Write 87 602 correct to 1 s.f.
3. Write 3738 correct to 3 s.f.
4. Write 0.439 correct to 2 s.f.
5. Write 0.5057 correct to 3 s.f.
6. The speed of light is 299 792 458 m/s. Write this correct to 3 s.f.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 800
2. 90 000
3. 3740
4. 0.44
5. 0.506
6. 300 000 000 m/s

Examples 5 and 6 — Estimating

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (2 marks)

A rectangular field is 48.7 m long and 31.4 m wide.
Work out an estimate for the area of the field.

Example 6 (2 marks)

Work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$48.7 \approx 50 \quad 31.4 \approx 30$

Area $\approx 50 \times 30$

$= \mathbf{1500\ m^2}$

Tip: write down the rounded numbers.

Example 6

$2.11 \approx 2, \quad 58.9 \approx 60, \quad 8.859 \approx 9$

$$\frac{2^2 \times 60}{\sqrt{9}} = \frac{240}{3}$$

$= \mathbf{80}$

Tip: round to 1 s.f. first.

Task 3

PRACTICE · 9 MIN

GOLD

Round each number to 1 significant figure first

1 Estimate 31.4×48.7

(2 marks)

2 Estimate $\dfrac{39.8 \times 5.1}{0.49}$

(2 marks)

3 Estimate $\dfrac{612 \times 0.38}{19.7}$

(2 marks)

4 Estimate $\dfrac{\sqrt{99.2} \times 3.1}{0.52}$

(2 marks)

5 A shop sells 289 drinks at $1.95 each.

(a) Work out an estimate for the money taken.

(b) Is your estimate bigger or smaller than the exact answer? Give a reason.

(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1 $30 \times 50 = 1500$

2 $\frac{40 \times 5}{0.5} = 400$

3 $\frac{600 \times 0.4}{20} = 12$

4 $\frac{\sqrt{100} \times 3}{0.5} = \frac{30}{0.5} = 60$

5 (a) $300 \times \$2 = \$600$

(b) Bigger, because both numbers were rounded up.

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write 6.284 correct to 1 d.p.

SILVER Write 0.00716 correct to 2 s.f.

GOLD Estimate $\dfrac{7.9 \times 41.2}{0.198}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE 6.3

SILVER 0.0072

GOLD $\frac{8 \times 40}{0.2} = 1600$

Exam Practice — Decimal Places

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

(a) Write 7.3862 to 2 d.p. (1)

.....

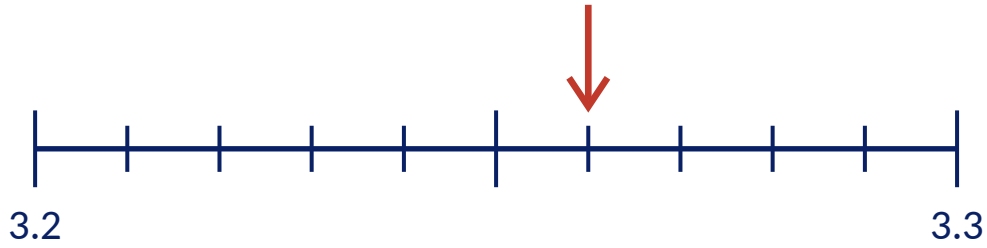
(b) Write 7.3862 to 1 d.p. (1)

.....

Question 2

(2 marks)

The arrow points to a number.



(a) Write down this number. (1)

.....

(b) Round it to 1 decimal place. (1)

.....

Exam Practice — Significant Figures

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

(a) Write 45 682 correct to 2 significant figures. (1)

.....

(b) Write 0.006047 correct to 3 significant figures. (1)

.....

Question 4

(2 marks)

The population of a town is 238 517.

(a) Write 238 517 correct to 1 significant figure. (1)

.....

(b) Write 238 517 correct to 3 significant figures. (1)

.....

Exam Practice — Estimating

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

The diagram shows a rectangular garden.

Turf costs £4.85 per m^2.

Estimate the cost of turf for the garden.

Diagram NOT accurately drawn



..... £

Question 6

(3 marks)

(a) Work out an estimate for

$$\frac{6.12 \times 39.8}{0.198}$$

(2)

.....

(b) Use your calculator to find the exact value. Was your estimate close?

(1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 7.39 B1

(b) 7.4 B1

Tip: look at the next digit only.

Question 2

(a) each small step = 0.01

3.26 B1

(b) 3.3 B1

Tip: work out what one small step is worth.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 46 000 B1

(b) 0.00605 B1

Tip: zeros at the start do not count.

Question 4

(a) 200 000 B1

(b) 239 000 B1

Tip: keep the number the same size.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$20 \times 8 = 160\ m^2$ — M1

160×5 — M1

= £800 — A1

Tip: round every number to 1 s.f. first.

Question 6

(a) $6 \times 40 \div 0.2$ — M1

= 1200 — A1

(b) 1230.18… (yes, close) — B1

Tip: dividing by 0.2 is the same as × 5.